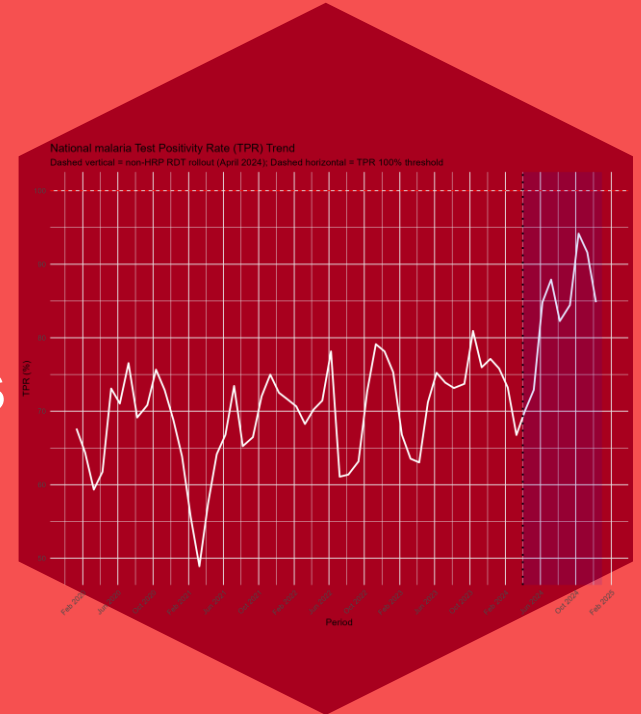


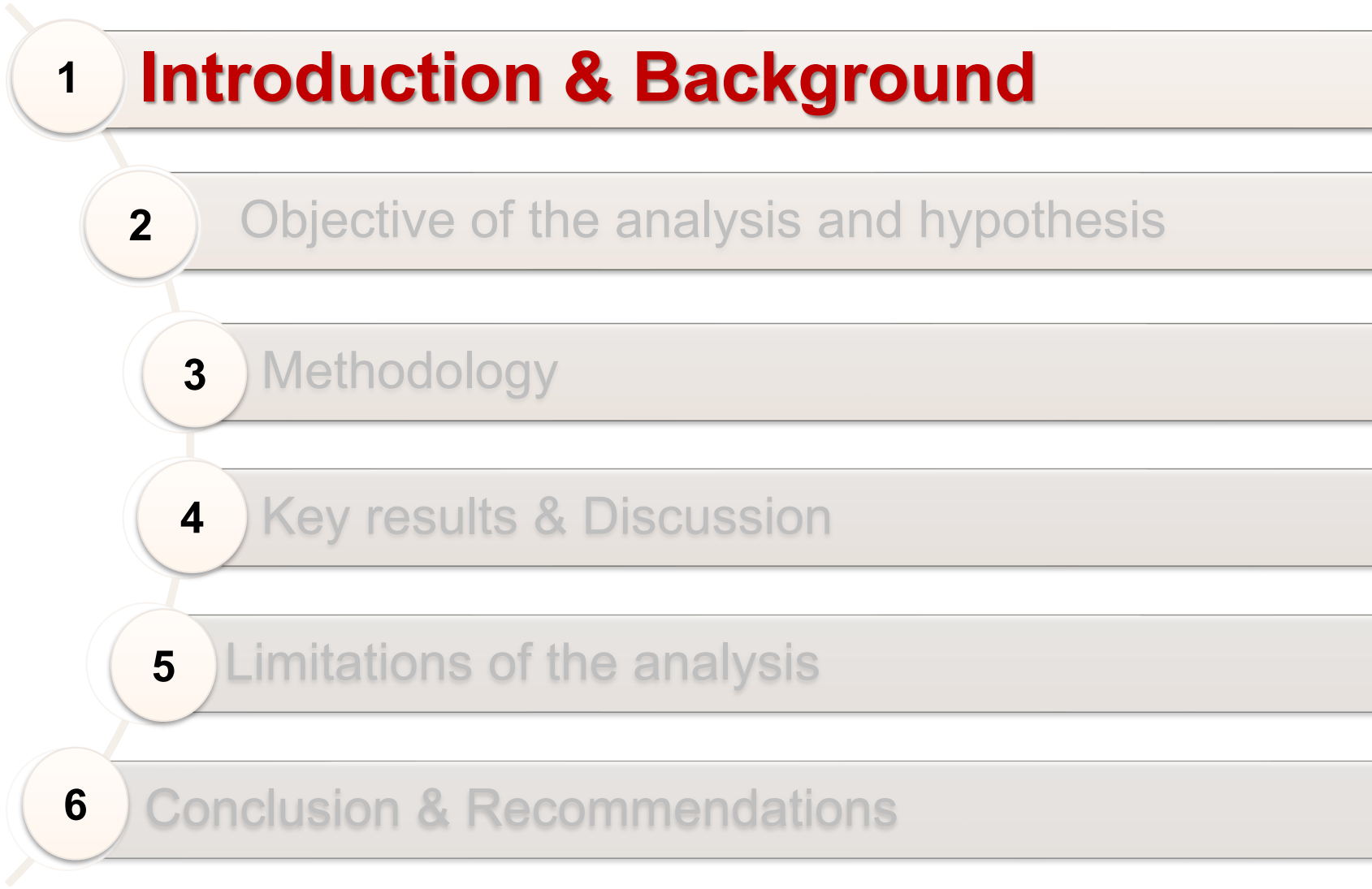
Malaria Case Trends in Ethiopia: Analysis Before and After the Rollout of Non-HRP RDTs

By: Enku Demssie

Data analytics fellow-Ethiopia

June 2025





1 Introduction & Background

2 Objective of the analysis and hypothesis

3 Methodology

4 Key results & Discussion

5 Limitations of the analysis

6 Conclusion & Recommendations

Introduction: *Malaria in Ethiopia*

- ~75% of Ethiopia is malarious; 69% of the population is at risk of malaria.
- In 2024, over 10.2 million malaria cases and 1,343 deaths were reported — a 6-fold increase from 2021/22 (~1.6 million cases, 180 deaths).
- Despite two decades of significant progress through LLINs, IRS, early diagnosis, and prompt treatment, a malaria resurgence has been observed since 2022.
- Multiple contributing factors are suspected, including:
 - HRP2/3 gene deletions, Emergence of *Anopheles stephensi*, Insecticide resistance, & Parasite resistance
 - Climatic and environmental changes,
 - Health system and programmatic challenges (disruption, intervention coverage gaps)
 - Social and demographic dynamics (pop displacement, migration, urbanization).

Background: *Why Confirmatory Diagnosis Matters in Malaria Control?*

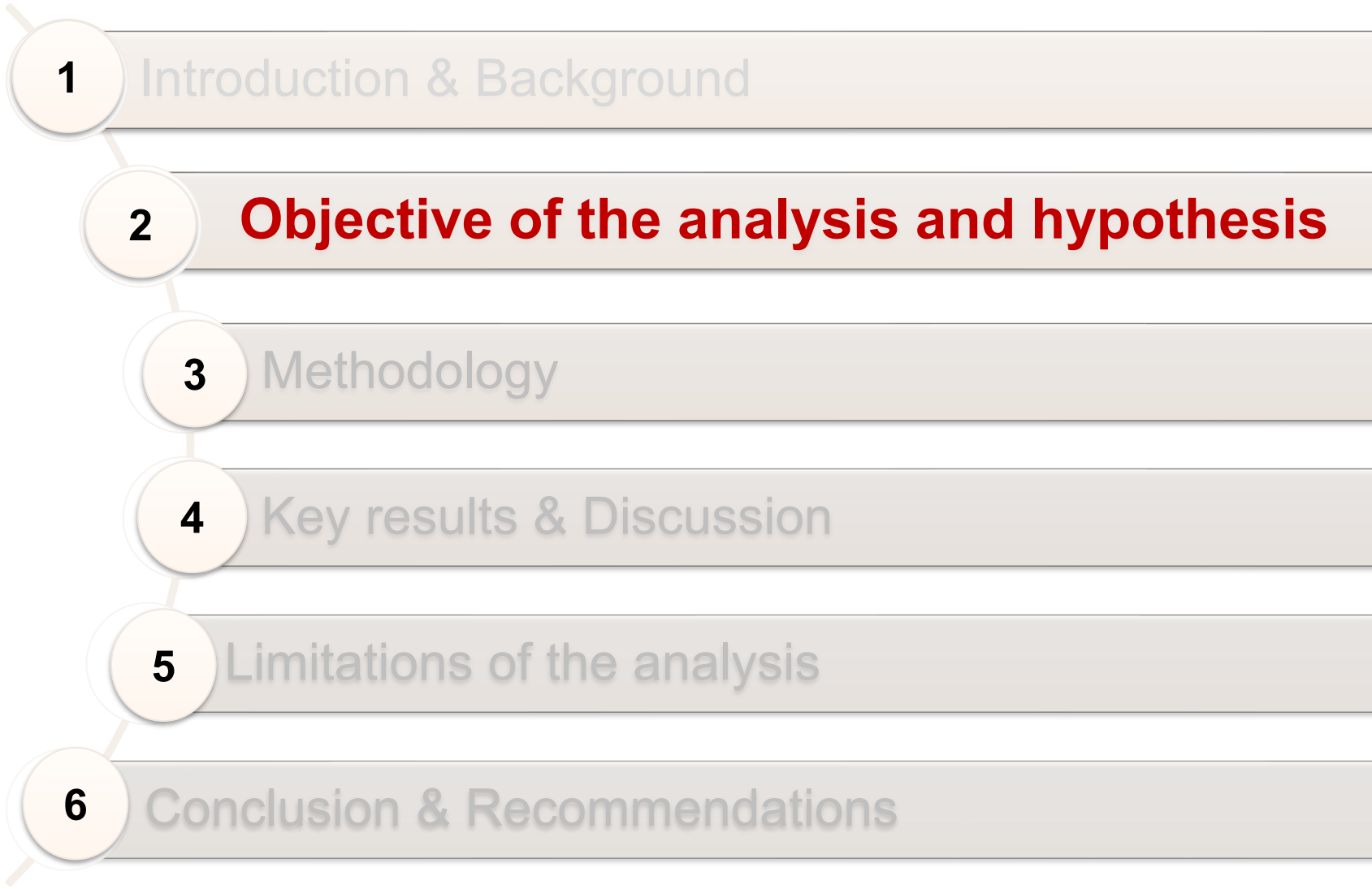
- Malaria often presents as a nonspecific febrile illness, clinically indistinguishable from other infections.
- WHO recommends parasitological confirmation before treatment.
- Accurate diagnosis is essential for:
 - Appropriate case management
 - Reducing the overuse of antimalarials
 - Tracking and interrupting transmission
- Two main diagnostic methods:
 -  RDTs – fast, simple, used widely at a peripheral level
 -  Light Microscopy – practical gold standard

Background: *RDTs: The Backbone of Malaria Diagnosis at Health Posts*

- RDTs are the primary tool for diagnosis at the health post level in Ethiopia.
- RDT antigen targets:
 - *HRP2* → *Pf-specific*
 - *Pf-pLDH* → *Pf-specific*
 - *Pv-pLDH* → *Pv-specific*
 - *Pan-pLDH* → *all species*
 - *Aldolase* → *all species*

Background: *HRP2/3 Gene Deletion & Diagnostic Shift*

- HRP2/3 gene deletions → false-negative RDT results for *P. falciparum*
- First reported in 2010; now seen in 41 endemic countries, including Ethiopia
- In Ethiopia (*Northern & Northwestern areas*):
 - HRP2/3 deletion prevalence: 8–15%
 - Crosses the 5% WHO threshold for diagnostic policy change
- MOH response:
 - 2023: Pilot of Pf-pLDH /Pv-pLDH combo RDTs in 4 districts
 - 2024: Nationwide rollout of non-HRP RDTs



1 Introduction & Background

2 **Objective of the analysis and hypothesis**

3 Methodology

4 Key results & Discussion

5 Limitations of the analysis

6 Conclusion & Recommendations

Objective of the analysis

General Objective

- To examine malaria case detection trends in Ethiopia before and after the nationwide rollout of non-HRP RDTs, using health post-level HMIS data from 2018 to 2024.

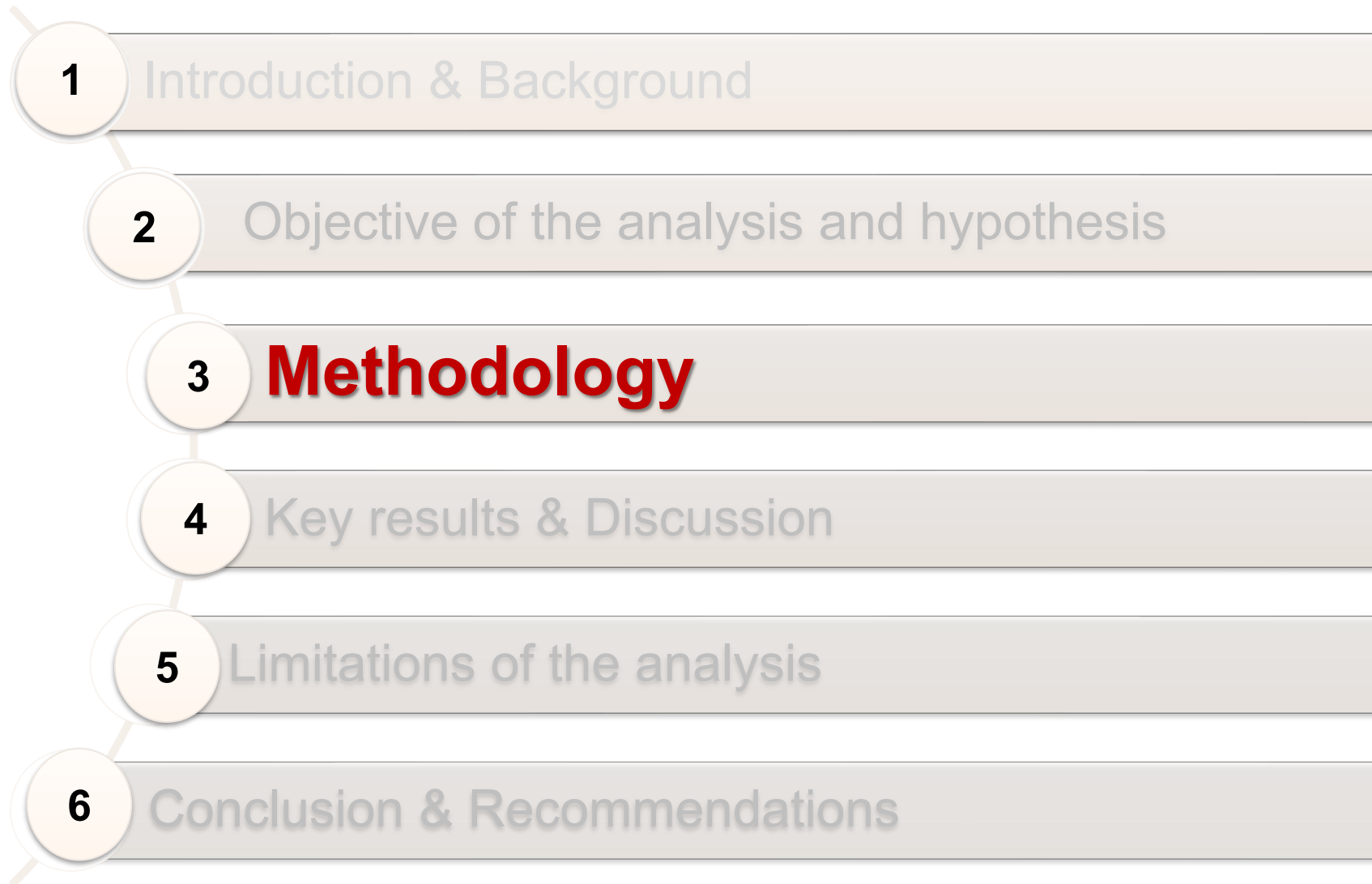
Specific Objectives

- To describe trends in malaria case reporting at health posts between 2018 and 2024.
- To compare malaria case detection patterns pre- and post-rollout of non-HRP RDTs.

Hypothesis

The nationwide rollout of non-HRP RDTs in April 2024 is hypothesized to:

1. Increase confirmed *Plasmodium falciparum* cases, particularly in regions with high HRP2/3 gene deletion prevalence.
2. Raise the Pf:Pv case ratio by improving the detection of Pf cases previously missed due to HRP2 gene deletions.
3. Elevate the test positivity rate (TPR), reflecting enhanced diagnostic sensitivity for Pf with non-HRP2 RDTs.



Methodology

Data Source

- National malaria data (Jan, 2020 – Dec, 2024), extracted from the HMIS (DHIS2) platform

Data Elements

- Number of malaria RDTs performed
- Confirmed malaria cases (positive tests)
- Species-specific diagnoses (from ICD classification)

Data Cleaning & Preparation

- Geographic consistency checks
- Harmonization of data elements
- Classification of HFs by their type

Analytical Approach

- Descriptive analysis comparing trends pre- and post-rollout of non-HRP RDTs (April 2024)

Visualization Techniques

- *Time series plots* – to show national and subnational malaria trends
- *Bar charts* – to illustrate health posts' contribution to national case counts
- *Heatmap* – to highlight data quality gaps, specifically:
 - ✓ Mismatch between total positive RDT tests and species-specific case count

Cont....

HP Selection Criteria

- Pre-rollout period (Jan 1st 2020 – Mar 31st 2024):
 - ≥ 25 confirmed malaria cases (*Pf* or test positive)
 - Reported data for $\geq \frac{2}{3}$ of months
- Post-rollout period (April 1st 2024 – Dec 31st 2024):
 - ≥ 5 confirmed malaria cases
 - Reported data for $\geq \frac{2}{3}$ of months
- Met criteria for both pre- and post-rollout periods

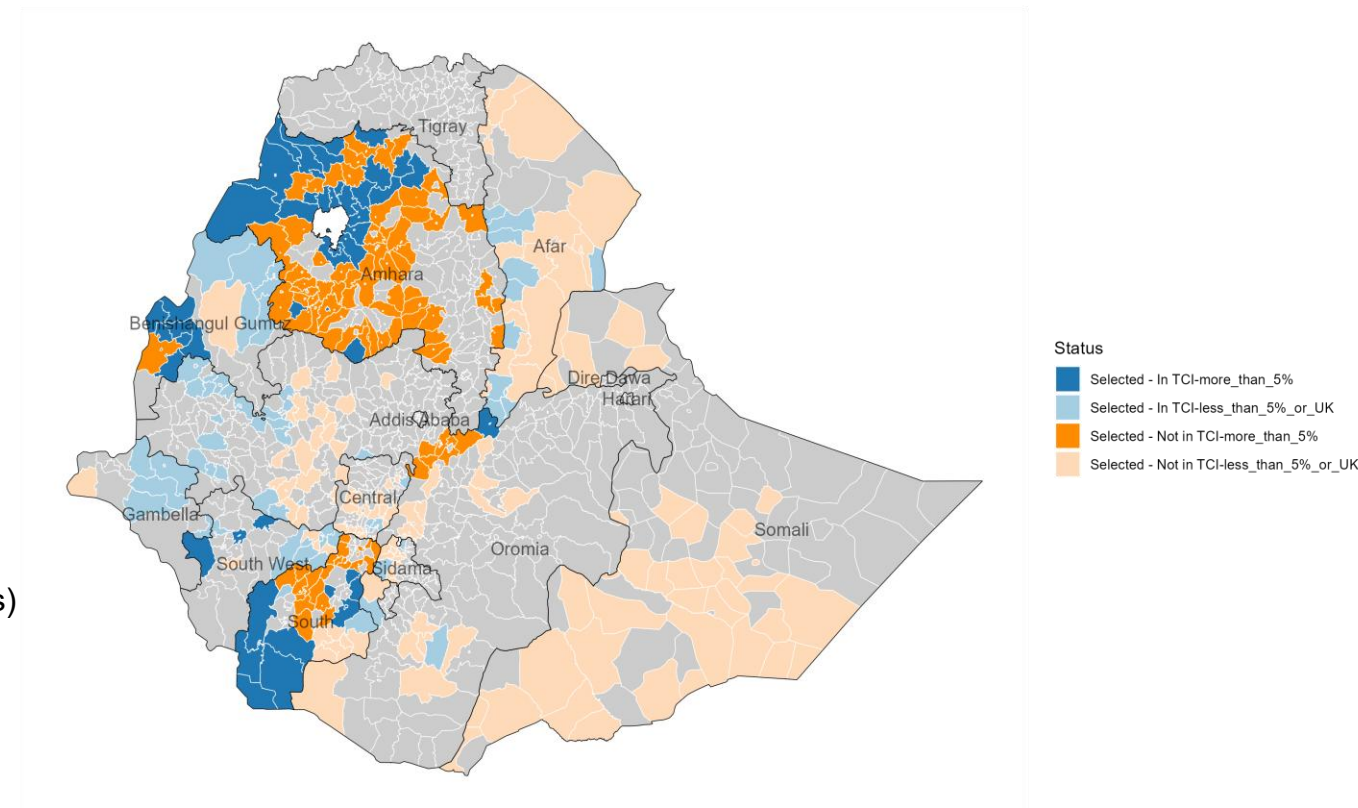
2,036 HPs selected from 333 districts (out of 18331 HPs)

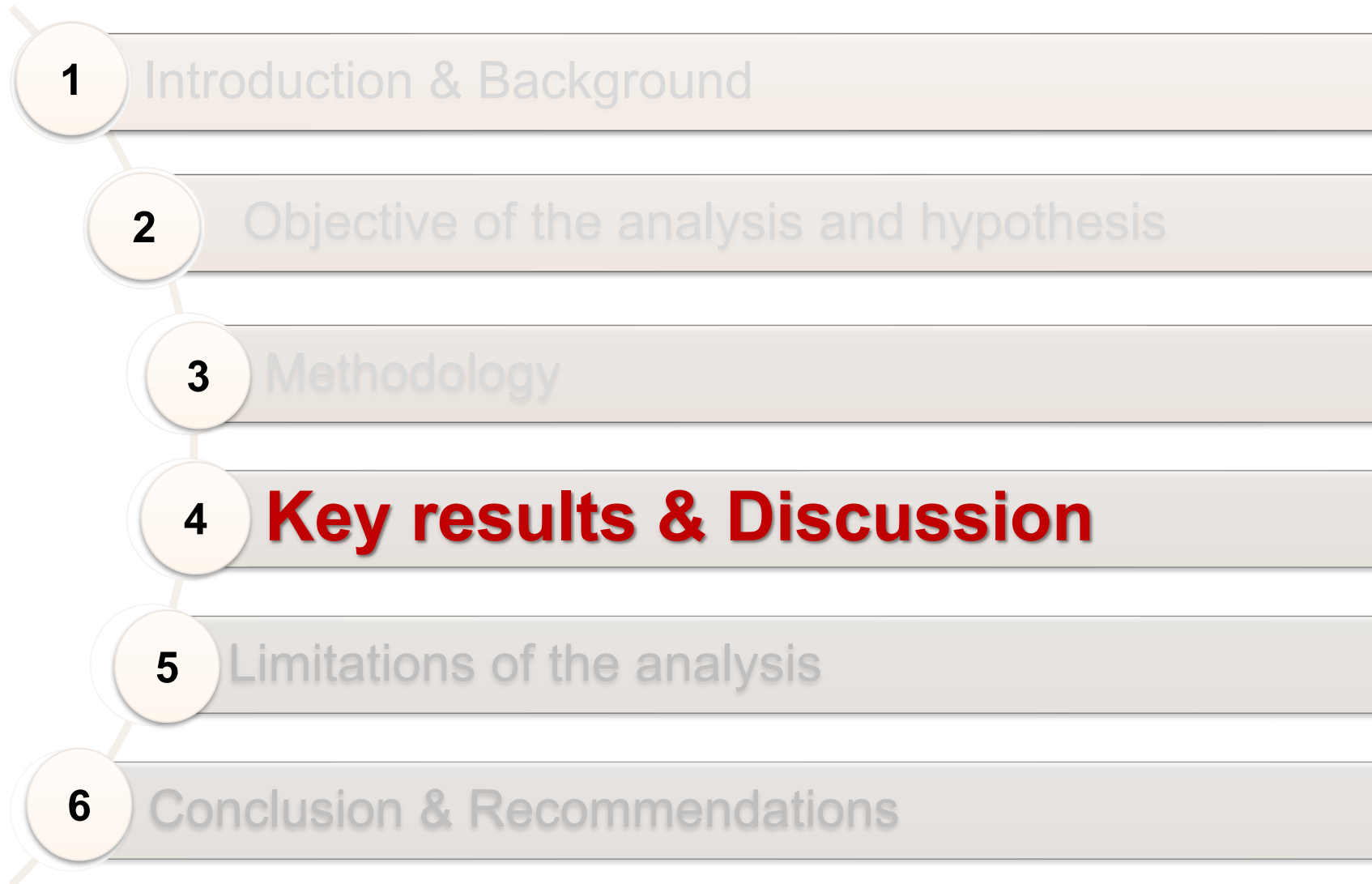
- 114 (34%) of the districts are part of the targeted Cluster Initiative & 140 (42%) have high HRP2/3 deletion prevalence

Figure 1: Geographic distribution of HPs selected for this analysis

Selected HFs by districts and the respective HRP2/3 Gene Deletion Prevalence and Targeted Cluster Initiative Membership status

TCI = Targeted Cluster Initiative, UK= Unknown,
5% = HRP2/3 gene deletion prevalence

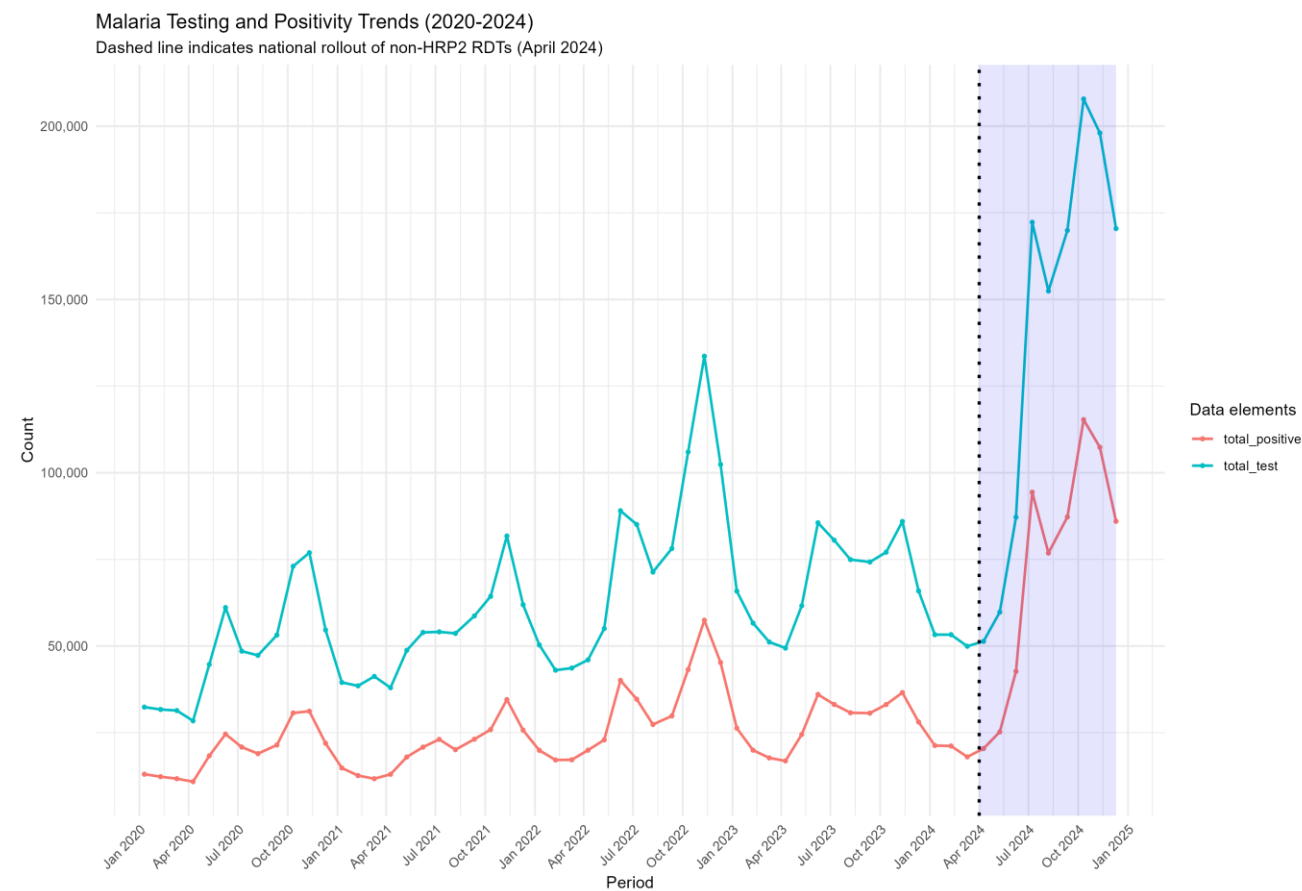




Trends in Malaria Testing and Positivity

- ~ 4.4 million malaria rapid diagnostic tests (RDTs) were conducted at the health posts included in this analysis during the study period.
- The annual number of tests rose from 583,248 in 2020 to 1,425,779 in 2024 (~2.5 fold) while confirmed positive cases rose from 235,894 to 715,932 (3x).
- Two key factors may drive the sharp rise in 2024:
 - The national rollout of non-HRP RDTs in April 2024.
 - The launch of the targeted cluster initiative approach in June 2024, which likely intensified case finding and surveillance efforts in response to the resurgence.

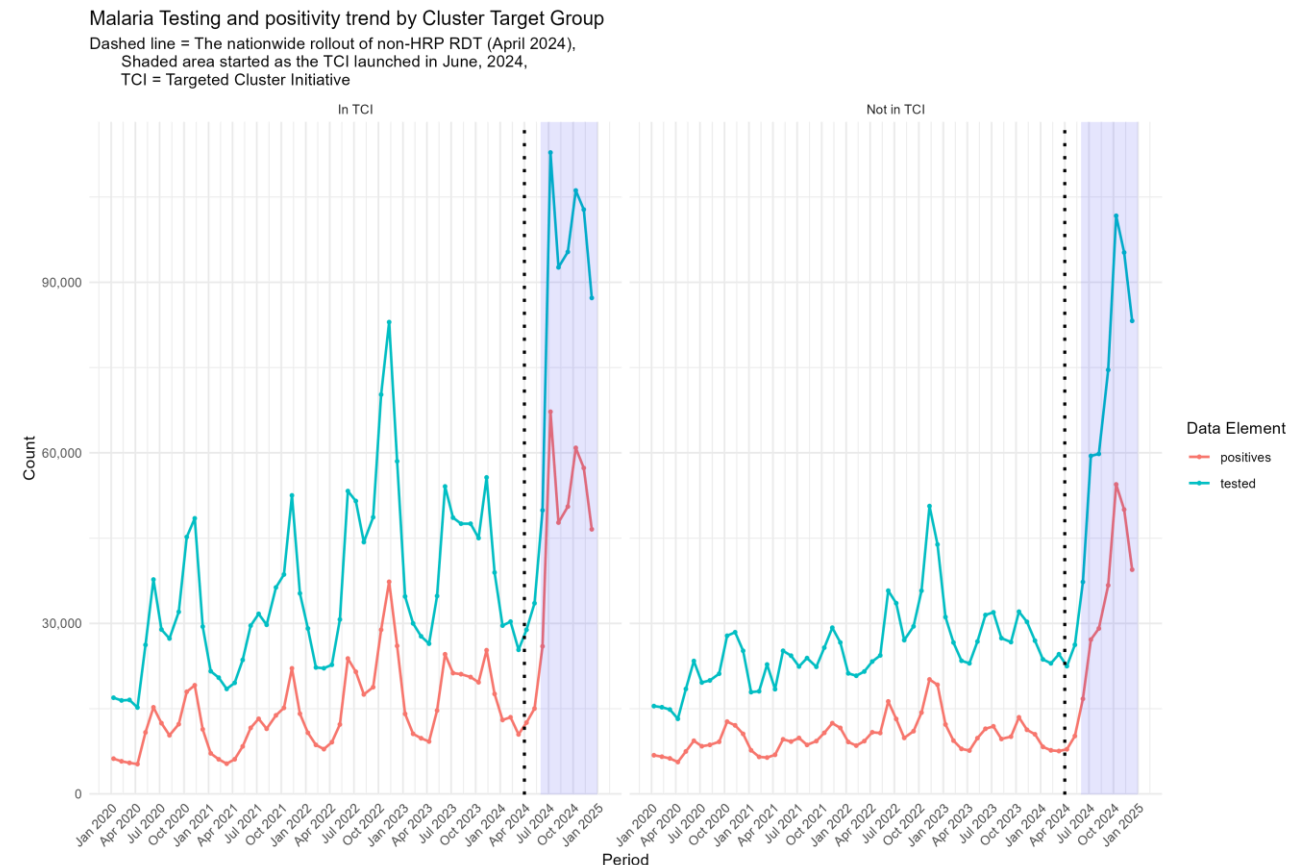
Figure 2: National malaria testing volume and positivity trend (2020–2024)



Testing & Positivity trend: Targeted vs. Non-targeted districts

- Both TCI targeted and non-targeted districts showed increased testing and confirmed malaria cases after the rollout.
- TCI-targeted districts:
 - Sharp rise peaking in July, soon after TCI launch.
 - Temporary dip in August, followed by a second peak in October.
 - Decline began in November, continuing through year-end.
- Non-targeted districts:
 - Gradual increase, peaking in October.
 - Smoother trend with no sharp fluctuations.
- In both groups, decline began before the end of the major transmission season (Dec).

Figure 3: Malaria Testing and Positivity trend by Cluster Initiative activity



Regional Trends in Malaria Testing and Positivity

- A notable increase in testing volume and positivity was observed across most regions following the rollout of non-HRP RDTs.
- The post-rollout peak in both testing and confirmed cases exceeded all previous levels during the analysis period.
- Two key factors may drive this increasing trend:
 1. Improved detection of HRP2/3-deleted *P. falciparum* and greater uptake of testing, which enhances diagnostic coverage.
 2. The implementation of the targeted cluster initiative approach has led to intensified surveillance efforts during the ongoing resurgence of malaria.

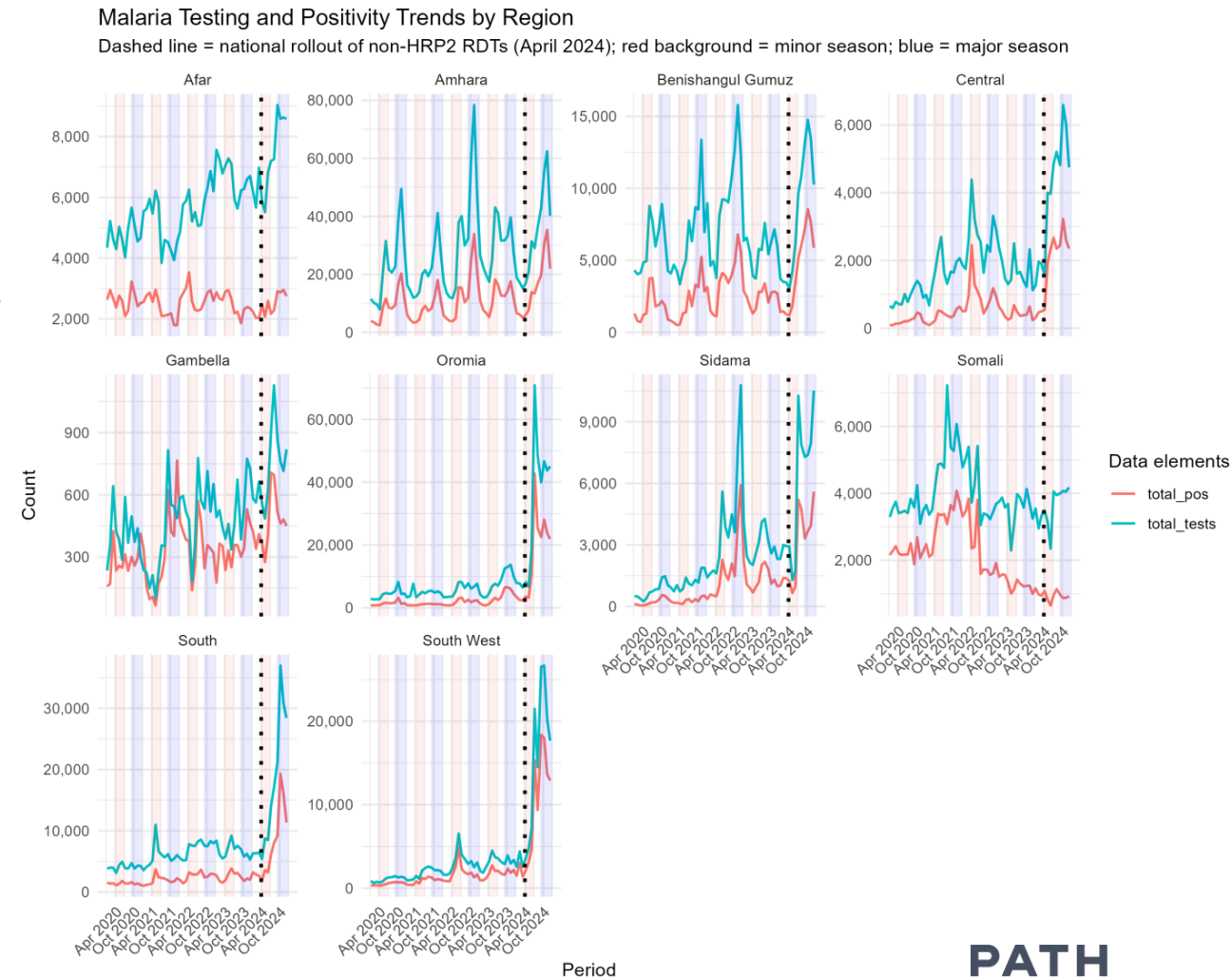
Figure 4: Regional malaria testing volume and positivity trend



Regional Trends cont....

- In the Somali Region, there was a decline following the rollout, but it then showed a modest rebound and reached stabilization.
 - Genuinely low transmission levels - a lower malaria burden in the eastern regions of Ethiopia.
 - Gaps in testing or reporting,
 - Highlighting the need for further investigation to understand the true drivers of this trend.
- Oromia, South, and South West regions experienced stronger increases in both testing and confirmed cases than before the rollout.
 - Intensified humanitarian response (west Oromia)
 - Renewed surveillance efforts
- Despite regional variation, trends were broadly aligned with Ethiopia's malaria transmission seasons.

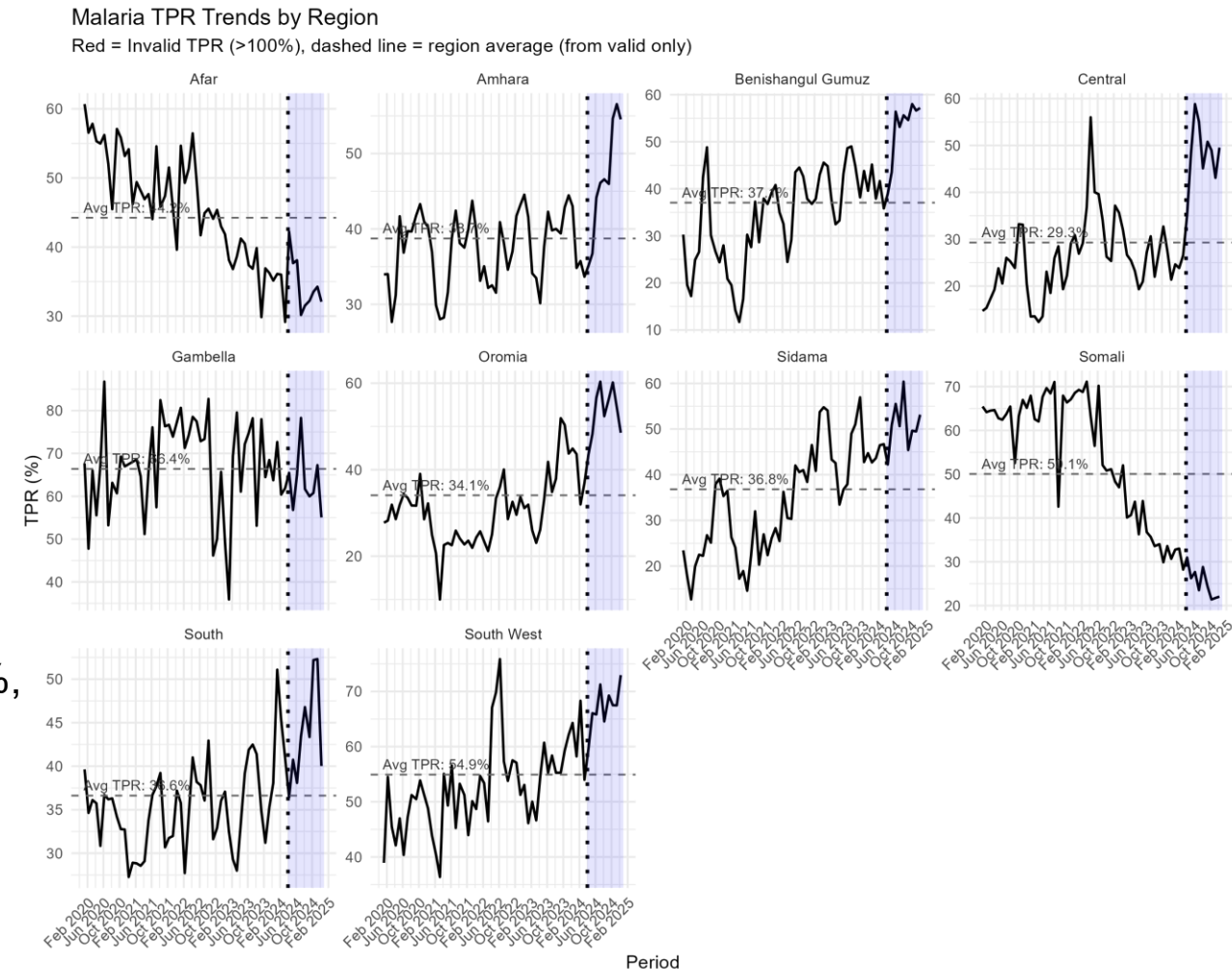
Figure 4: Regional malaria testing volume and positivity rate (2020–2024)



TPR: Seasonal & Regional Trends

- TPR exhibited seasonal peaks aligned with Ethiopia's major and minor malaria transmission seasons.
- Following the nationwide rollout of non-HRP RDTs, TPR increased notably in Amhara, Benishangul-Gumuz, Central, Oromia, and Southwest regions, while the Somali Region showed a consistent decline.
- Highest regional average TPRs were observed in Gambella, Southwest, and Somali (ranging from 50% to 66%).
- Central Region recorded the lowest average TPR at 29%, while the remaining regions ranged between 34% and 44%.

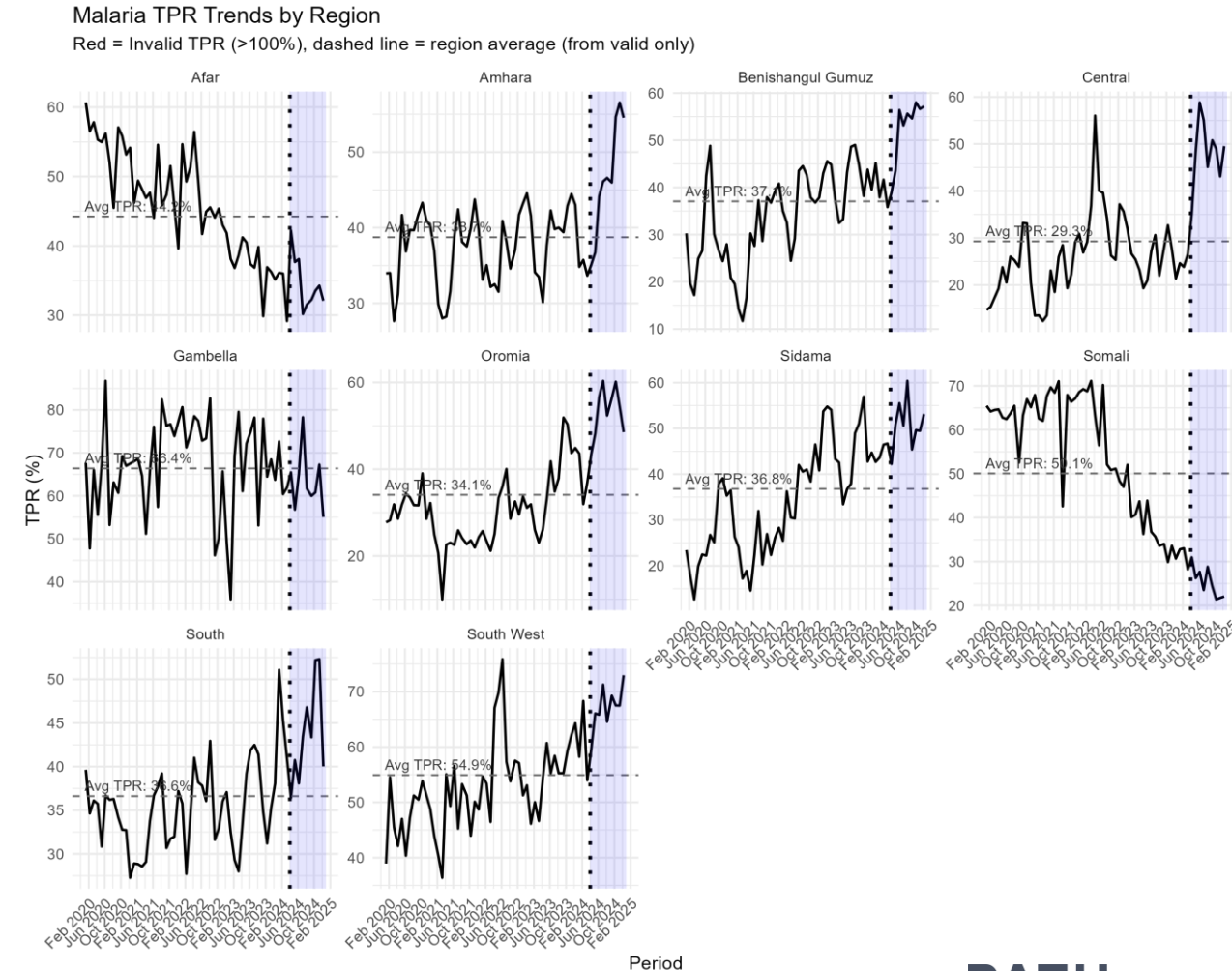
Figure 5: Regional test positivity rates monthly trend



Trends in TPR cont....

- The prominent increase in TPR in some of the regions after April 2024 likely reflects a confluence of factors:
 - Improved detection of HRP2/3 gene deleted pf species by the Pf-pLDH-based RDTs.
 - More targeted testing during the resurgence response.
 - Enhanced reporting practices linked to increased surveillance intensity.
 - Expanded community-level outreach testing by Health Extension Workers in some areas, increasing access and case identification in previously under-tested populations.
- However, this may also reflect overly selective testing practices, where some health posts may have missed cases that still met the case definition.

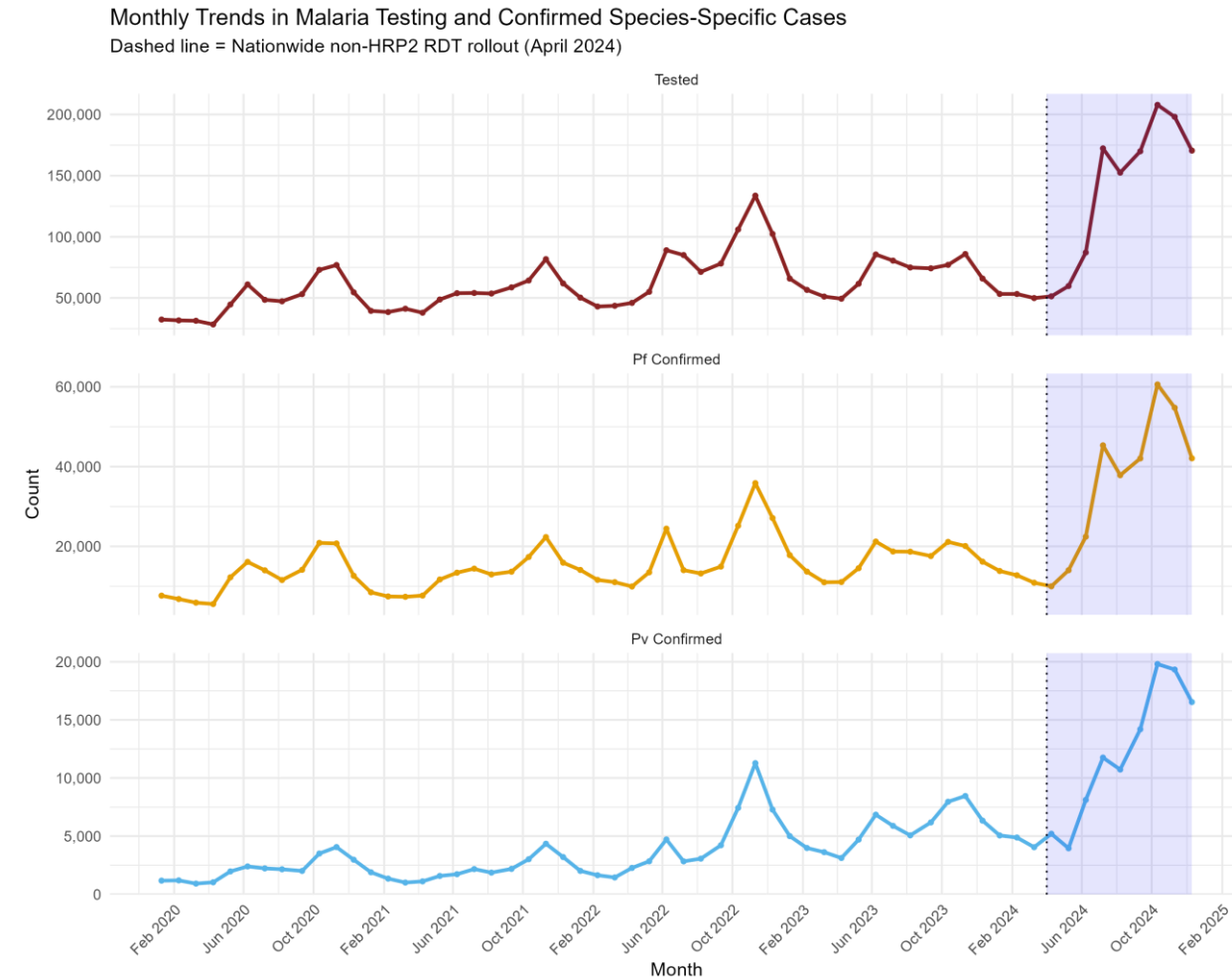
Figure 5: Regional test positivity rates monthly trend



Species specific trend

- *P. falciparum* accounted for ~78% and *P. vivax* for ~21% of confirmed malaria cases among selected health posts.
- From 2020 to 2024, proportion of *P. vivax* cases increased (15% → 25%), while *P. falciparum* declined (85% → 74%).
- After the April 2024 rollout of non-HRP RDTs, confirmed cases for both species rose, peaking in October. After October, cases declined but remained above pre-rollout levels.
- A few *P. ovale/malariae* cases were reported (2022–2024), likely data entry errors- excluded from species-specific analysis.

Figure 6: *Pf* and *Pv* malaria case volume trend before and after the rollout of non-HRP RDT

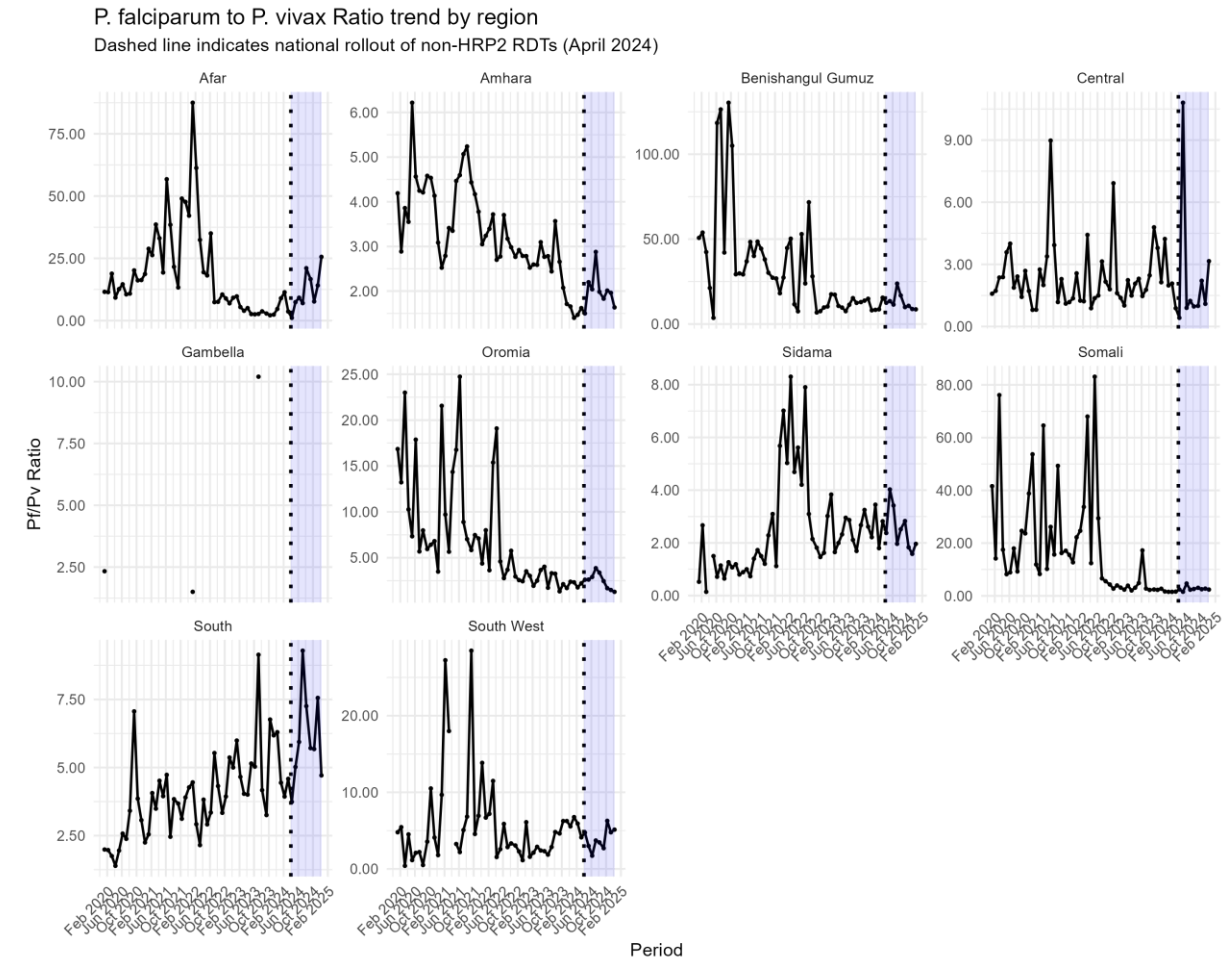


Species-specific trend cont...

Pf:Pv Ratio

- Following the rollout of non-HRP RDTs in April 2024, the Pf:Pv ratio fluctuated across most regions.
- In Afar, Sidama, and South Ethiopia, the ratio initially increased after the rollout but declined after July 2024, indicating a rising proportion of *P. vivax* cases.
- In Amhara and Benishangul-Gumuz, where some areas are known to have high HRP2/3 gene deletion prevalence, the Pf:Pv ratio saw an increasing trend after the rollout, which peaked modestly around July and then declined, suggesting a relative increase in *P. vivax*.
- Oromia exhibited a similar trend, with a decreasing ratio after July, further pointing to a growing presence of *P. vivax* in the post-rollout period.

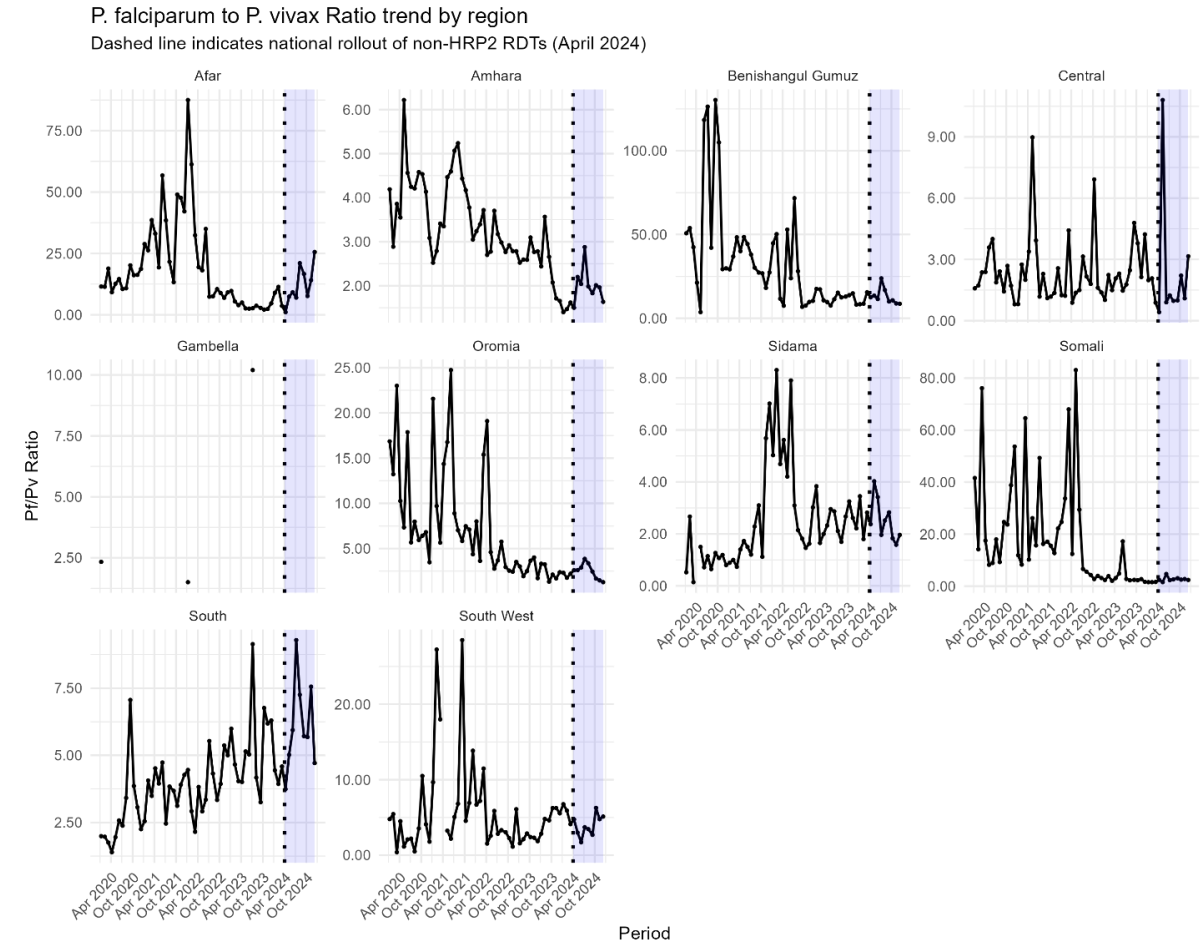
Figure 7: Malaria test positives and species confirmed cases trend by region



Species-specific trend cont.

- The post-rollout rise in the number of *P. falciparum* (Pf) cases likely reflects improved detection of HRP2/3-deleted strains following the shift to non-HRP RDTs, especially in areas with known gene deletions (*North & Northwestern*).
- The increasing proportion of *P. vivax* (Pv) cases is unlikely to result from the diagnostic change, as Pv-pLDH detection was already in place before April 2024.
- The Pv trend may instead point to a genuine shift in species dynamics, influenced by:
 - *Biological factors*: Pv relapse from dormant liver-stage hypnozoites.
 - *Treatment limitations*: Inadequate/failed provision and poor adherence to primaquine for radical cure
 - *Drug use patterns*: Use of AL instead of chloroquine in some areas.

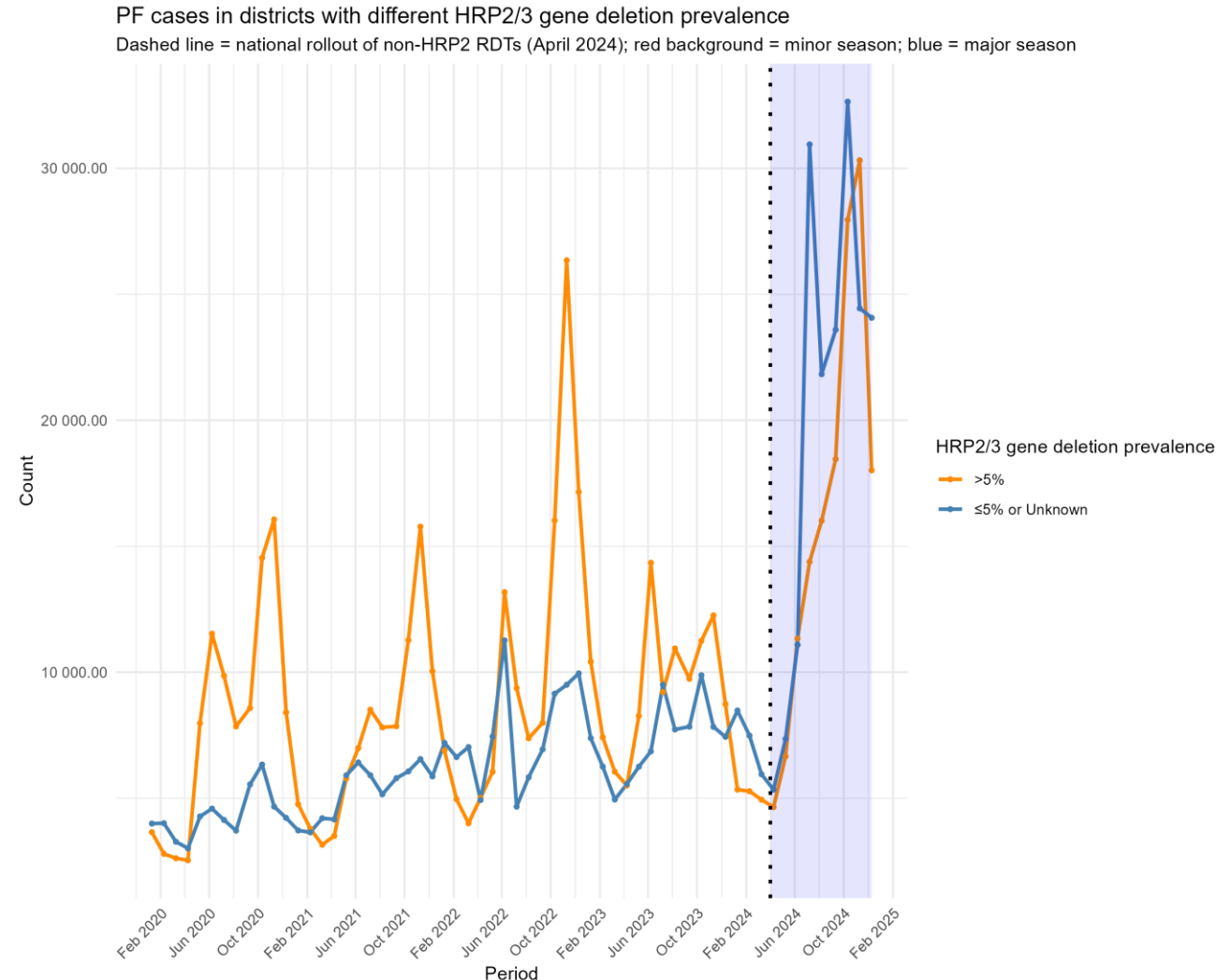
Figure 7: Malaria test positives and species confirmed cases trend by region



Species-specific trend cont.

- *Pre-rollout* → Both groups showed fluctuating trends. *Pf* case counts were consistently higher in low/unknown deletion areas compared to high-prevalence areas
- Post-rollout → Low/Unknown Prevalence Areas:
 - Steep rise with peaks in July and October, brief dip in August, and declining trend after Oct.
- Post-rollout → High Prevalence Areas:
 - Steady, uninterrupted rise peaking in November, followed by a decline.
- While a larger post-rollout rise was expected in high HRP2/3 deletion areas due to improved detection, the observed trend, particularly the greater increase in low/unknown areas, may reflect both **gaps in HRP2/3 prevalence data** and **the ongoing malaria resurgence** in Ethiopia, making it difficult to disentangle diagnostic from epidemiological drivers.

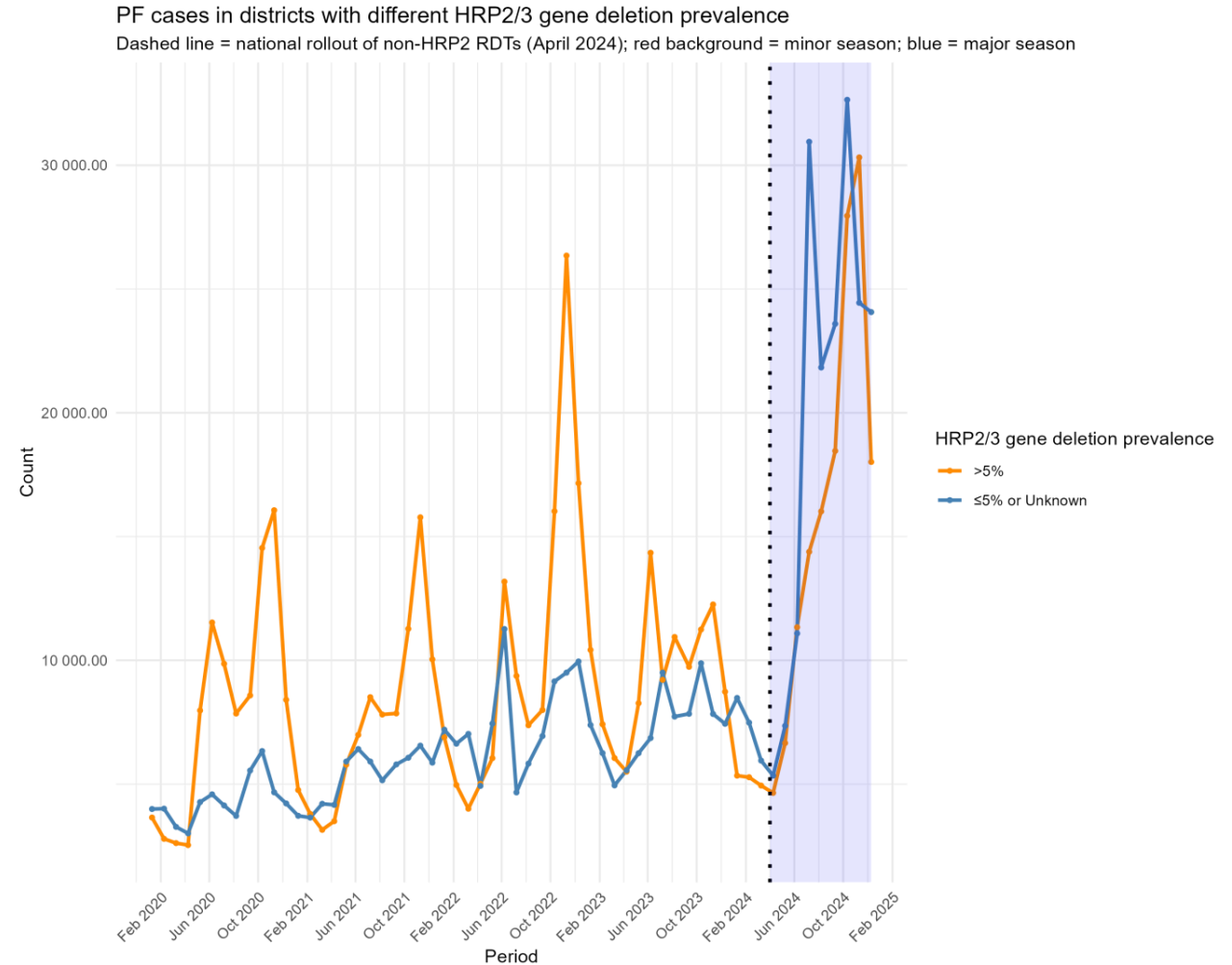
Figure 8: *Pf* cases trend by HRP2/3 gene deletion prevalence status



Species-specific trend cont.

- While a larger post-rollout rise was expected in high HRP2/3 deletion areas due to improved detection, the observed trend, particularly the greater increase in low/unknown areas, may reflect both **gaps in HRP2/3 prevalence data** and the **ongoing malaria resurgence in Ethiopia**, making it difficult to disentangle diagnostic from epidemiological drivers.

Figure 8: PF cases trend by HRP2/3 gene deletion prevalence status



Regional Shifts in Malaria Burden (2020–2024)

- Malaria case distribution showed significant regional variation among the selected health posts.
- Following the nationwide rollout of non-HRP2 RDTs in April 2024, most regions experienced a sharp rise in cases, with prominent peaks observed in July and October.
- The Amhara region consistently reported the highest number of confirmed cases, increasing from ~92,000 in 2020 to over 139,000 in 2024.
- Afar and Benishangul-Gumuz were major contributors until 2023, after which Oromia, South, and Southwest regions experienced substantial increases in 2024.
- The Somali region showed a distinct trend, with overall declining case numbers despite short-term fluctuations.
- Post-rollout peaks in the majority of the regions correspond with Ethiopia's major & minor malaria transmission seasons, reflecting expected seasonal patterns.

Figure 9: Regional contribution to the national positive cases



Regional Shifts cont...

- The ongoing rise in the number of cases in the Amhara region after the rollout of the non-HRP RDT could be due to:
 - 48% of the health posts selected from the region are part of the targeted cluster initiative, which may have led to increased case finding and reporting.
 - Improved detection of HRP2/3-deleted *P. falciparum* cases using non-HRP RDTs in areas with known high prevalence of gene deletions.
 - Catch-up effects following periods of conflict-related service disruption in the region, possibly inflating case numbers once health services resume.
- In Oromia, a sharp rise in malaria cases started in 2023, particularly after April 2024. This can be attributed to :
 - Increased transmission and enhanced detection of HRP2/3 gene-deleted strains.
 - Intensified humanitarian response and last-mile service delivery prompted by the ongoing conflict, particularly in western Oromia, which may have facilitated broader deployment and use of diagnostic tools.

Regional Shifts cont...

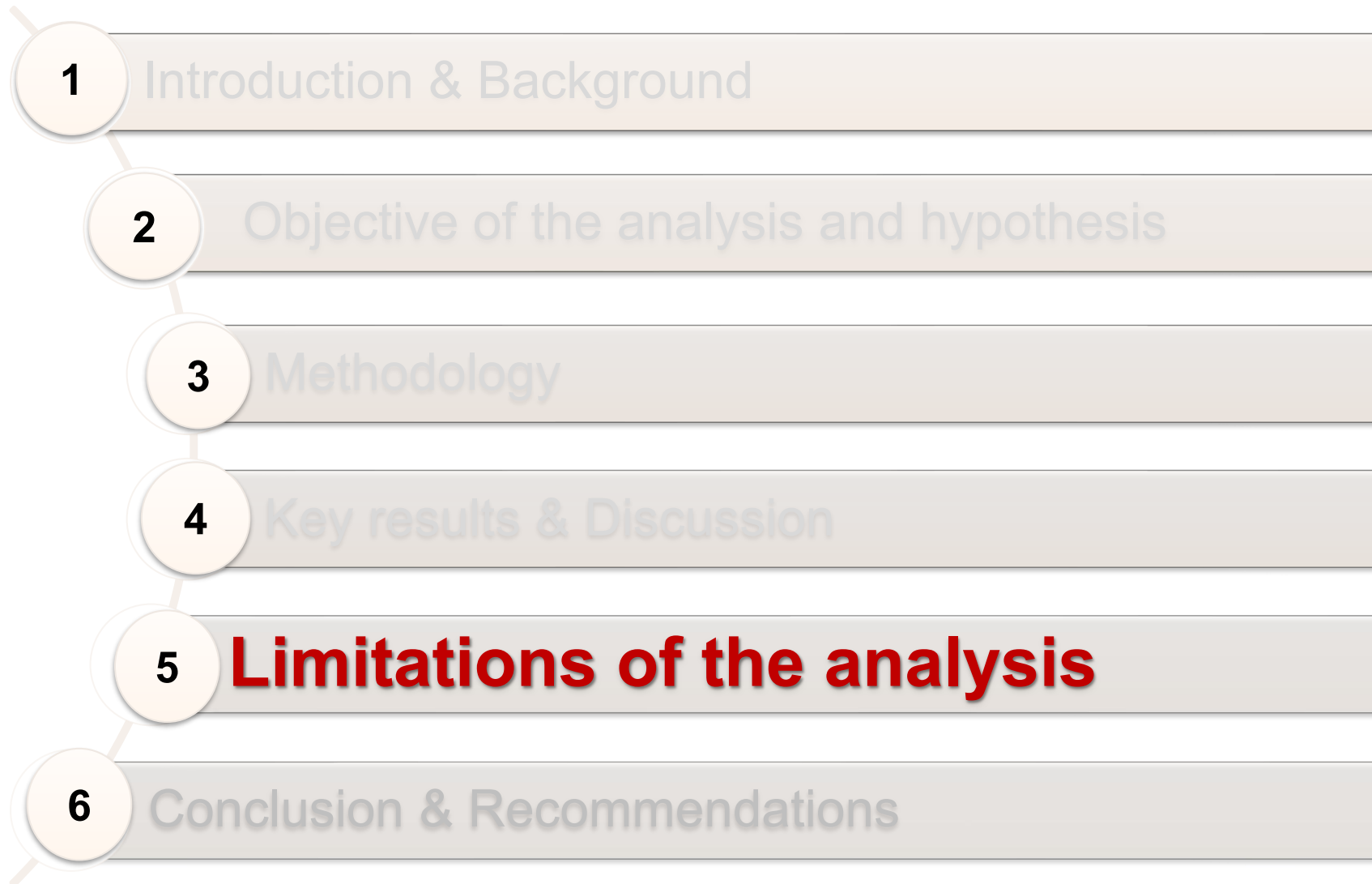
- **Benishangul-Gumuz:** A sharp rise in cases after the April rollout, peaking in October, may reflect improved detection of HRP2/3-deleted *P. falciparum* strains using non-HRP RDTs, as the region includes areas with suspected high gene deletion prevalence.
- **South, Southwest, and Central regions:** Post-rollout increases may be driven by enhanced detection of HRP2/3-deleted strains and strengthened surveillance efforts.
- The decline after the April rollout in the Somali region may be attributed to low transmission rates, gaps in RDT supply, or weakened health service delivery; further investigation is required.

Seasonal and Diagnostic Considerations

- Malaria cases typically peak during the major (Sep–Dec) and minor (Apr–Jun) transmission seasons across most regions.
- In 2024, an atypical and prolonged increase in confirmed malaria cases began in April and extended beyond the typical minor transmission season, with peaks observed in July across many regions.
- This may reflect a combination of increased transmission and improved case detection following the rollout of non-HRP RDTs.
- Expanded testing coverage and improved surveillance may have also contributed to the continued increase.

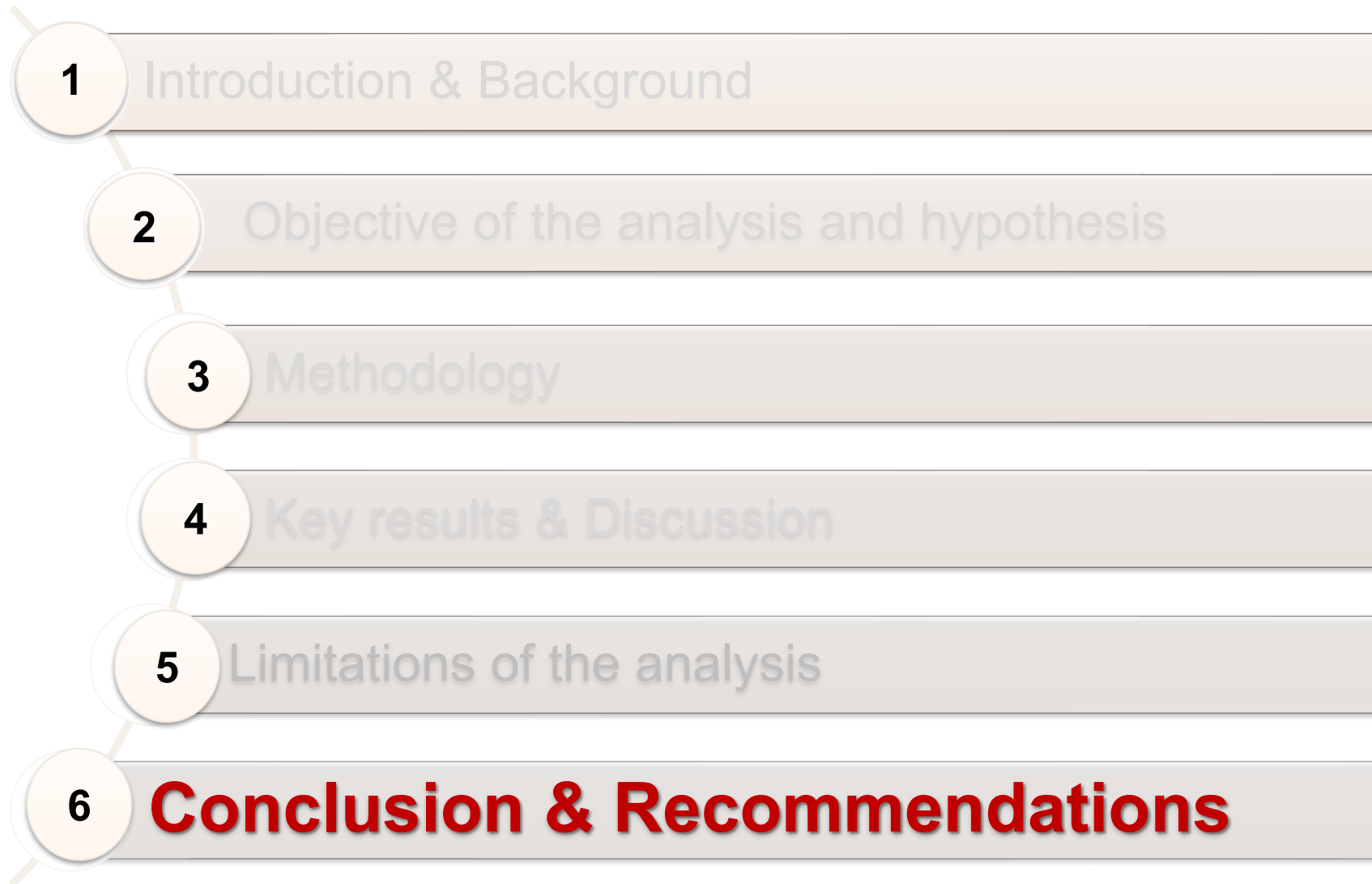
Figure 4: Regional malaria testing volume and positivity rate (2020–2024)





Limitations of the analysis

- *Incomplete reporting:* Data gaps, especially in conflict-affected areas, may lead to underrepresentation of cases.
- *Data quality issues:* Inconsistencies in confirmed case reporting (e.g., positives > tests) and species misclassification may affect trend accuracy. These records were retained to avoid substantial data loss.
- *Overlap with malaria resurgence:* The diagnostic transition occurred alongside a national upsurge in malaria cases, making it difficult to attribute the observed trends solely to the non-HRP RDT rollout.
- *RDT implementation variability:* Despite the national launch in April 2024, the actual switch to non-HRP RDTs varied by region; many sites continued using HRP-based RDTs until old stocks were exhausted.
- *Mixed use during transition:* Concurrent use of both HRP and non-HRP-based RDTs during the transition period may have affected trend consistency.
- *Non-standard facility naming:* Over 200 facilities were excluded due to inconsistent or unclear naming in the HMIS data, which made it difficult to reliably classify them by facility type.



Conclusion

- The sharp rise in cases starting in April 2024, with an initial peak in July followed by a higher peak in October, likely reflects a combination of factors, including intensified transmission, improved detection of HRP2/3-deleted *P. falciparum*, and heightened programmatic efforts through the Targeted Cluster Initiative across the included health posts.
- The proportion of *P. vivax* cases rose steadily, reaching 25% in 2024, despite the use of consistent diagnostic tools, suggesting possible species shift, relapse from inadequate radical cure, or declining treatment adherence. These trends warrant further investigation through focused research.
- Significant regional variation in trends highlights the need for context-specific surveillance and intervention strategies, tailored to local epidemiological patterns and operational realities.
- Observed peaks coincide with Ethiopia's known seasonal transmission periods, underscoring the continued importance of seasonal preparedness and timely implementation of vector control measures.

Recommendations

- Ensure diagnostic accuracy by maintaining rigorous quality assurance for non-HRP RDTs and monitoring their performance, especially in areas with suspected HRP2/3 deletions.
- Track species-specific trends to detect shifts in malaria epidemiology and guide tailored interventions, particularly in response to the rising proportion of *P. vivax* cases.
- Strengthen regional surveillance and response, with targeted support for high-burden, emerging transmission zones, and conflict-affected areas to improve timely case detection and reporting.
- Improve data quality and completeness, including standardizing geographies & health facility names, and enhancing granularity to enable more precise and actionable analysis.
- Evaluate the impact of diagnostic transitions through operational research to differentiate the effects of improved detection from actual changes in malaria transmission.
- Support sustained and timely vector control interventions aligned with seasonal trends and outbreak alerts to mitigate future surges in malaria burden.

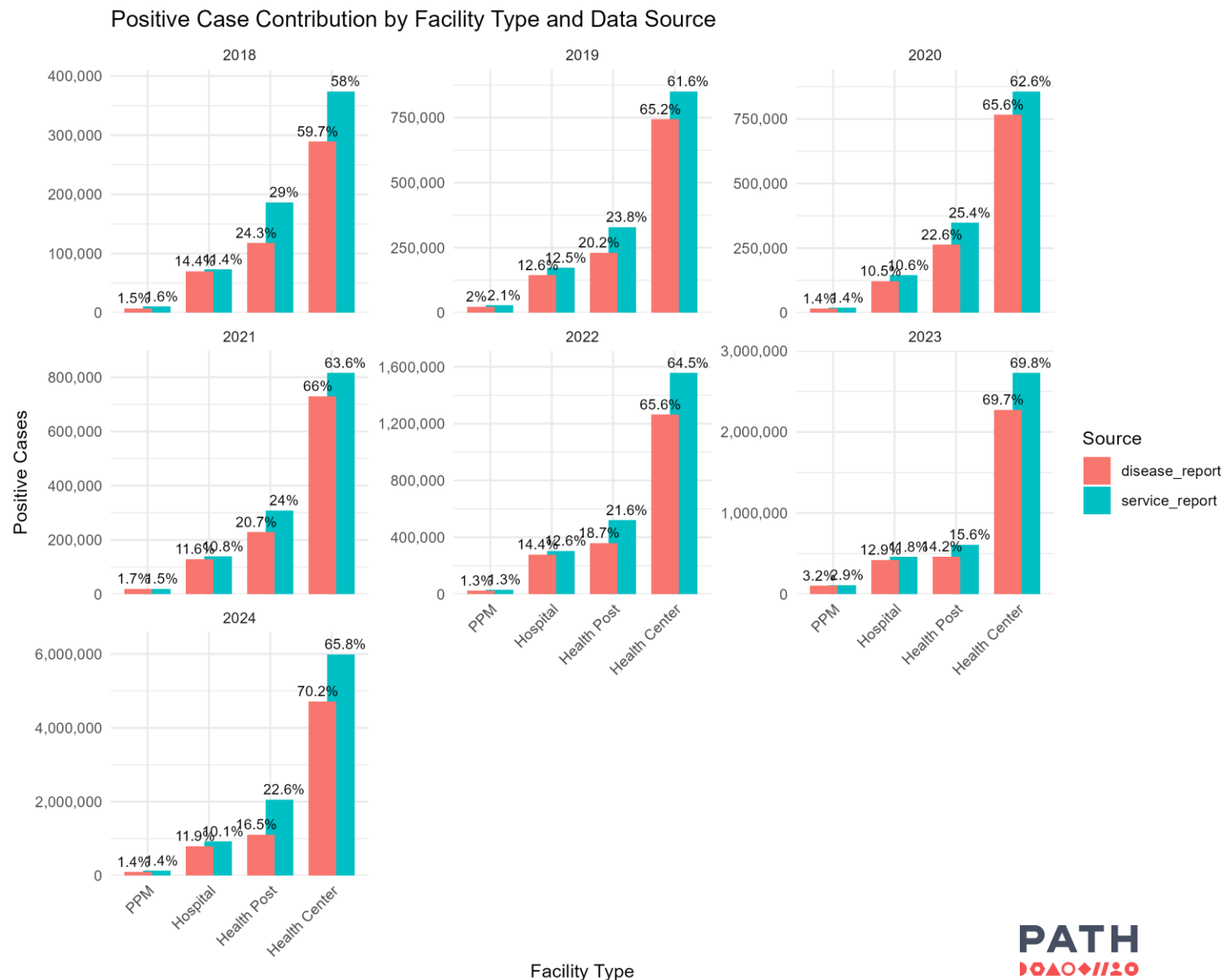
PATH
▷◈::▲○◆//人◻◉

1. HP contribution

Health post contribution to the overall case identification

- **Data source: National HMIS (all HFs)**
- **Note:** data from only health facilities that met consistent reporting criteria (at least 9/12 monthly reports each year) were used.
- The contribution of health posts to malaria case detection declined steadily from 2018 to 2023, reaching its lowest point in 2023.
- **A notable rebound was observed in 2024**, which could be driven by the malaria resurgence, which may have prompted the reactivation of previously underutilized or inactive health posts, particularly in severely affected areas.

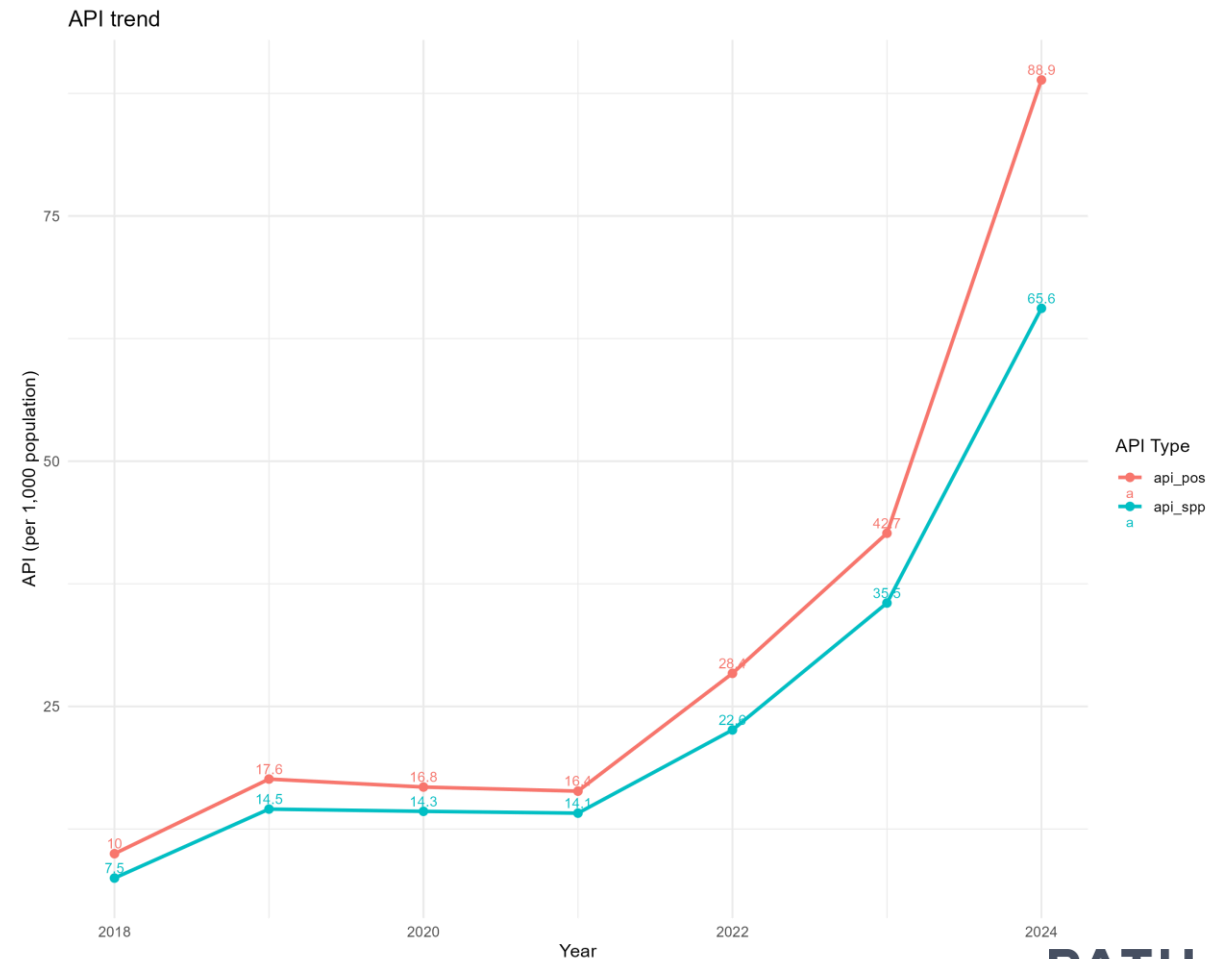
Figure 10: Health posts' contribution to the national case notification



2. Malaria Incidence Rate

- **Data source: National HMIS (all HFs)**
- **Note:** The national API values presented are calculated using data from health facilities that met consistent reporting criteria (at least 9/12 monthly reports each year) within each woreda. Consequently, not all facilities or woredas may be fully represented, potentially leading to **underestimation or uneven representation** of malaria incidence across the country.

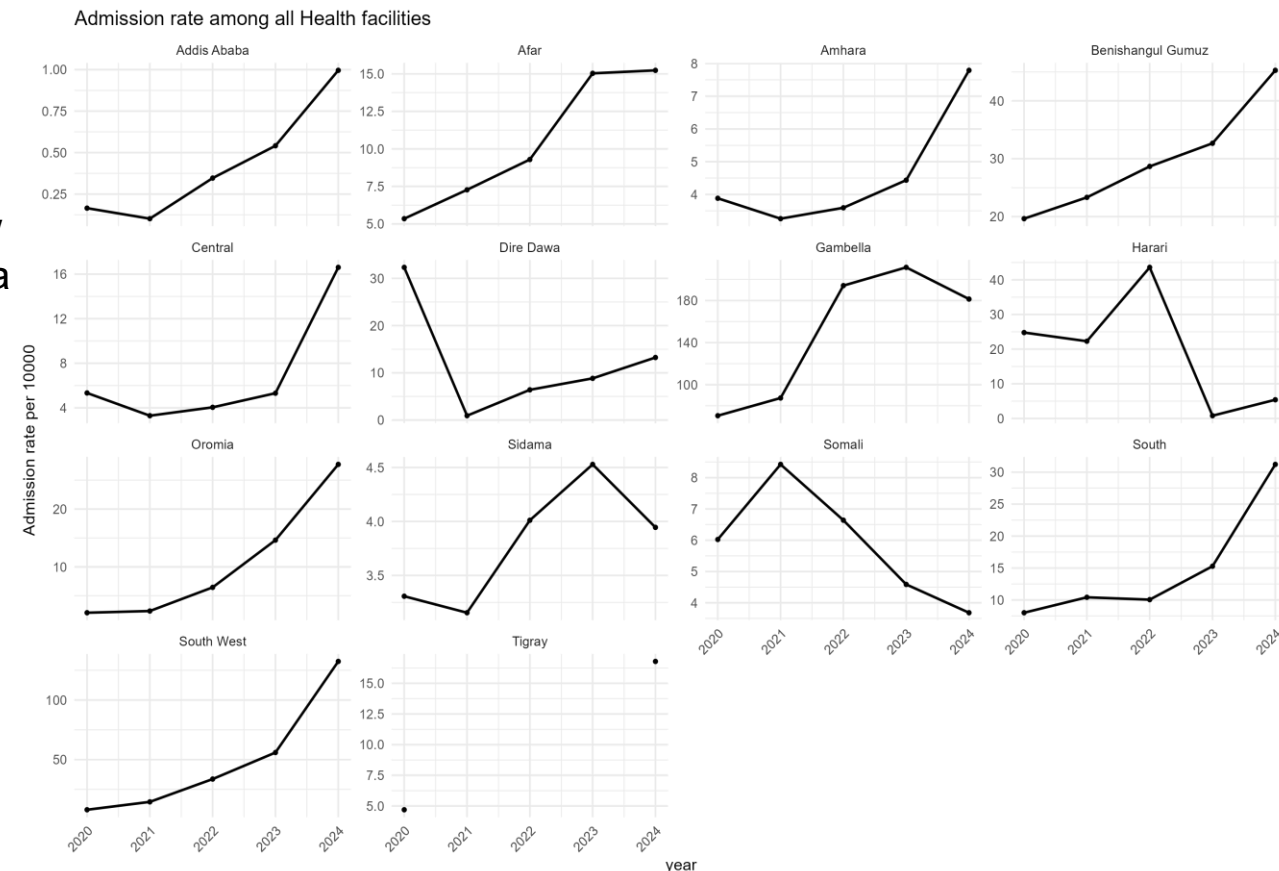
Figure 11: National malaria incidence rate comparison (positive tests vs species-specific report sums as numerators)



3: Malaria case admission rate

- **Data source: Hospital-level HMIS report**
- Conducted using HMIS data from hospitals that fulfill the consistently reporting facility criteria (9/12)
- Aimed to explore whether improved diagnostic accuracy for HRP2/3 gene-deleted *Pf* strains at HPs, enabled by the new RDTs, may have contributed to a reduction in severe malaria cases requiring hospitalization.
- Overall, malaria admissions increased in most regions following the rollout of the non-HRP RDT, despite some fluctuations.
- The post-rollout period was generally marked by higher admission levels compared to the pre-rollout period.
- Conclusion: While enhanced diagnosis may have supported earlier detection and treatment in some areas, other factors likely influenced the overall admission rate patterns. Warranted further investigation

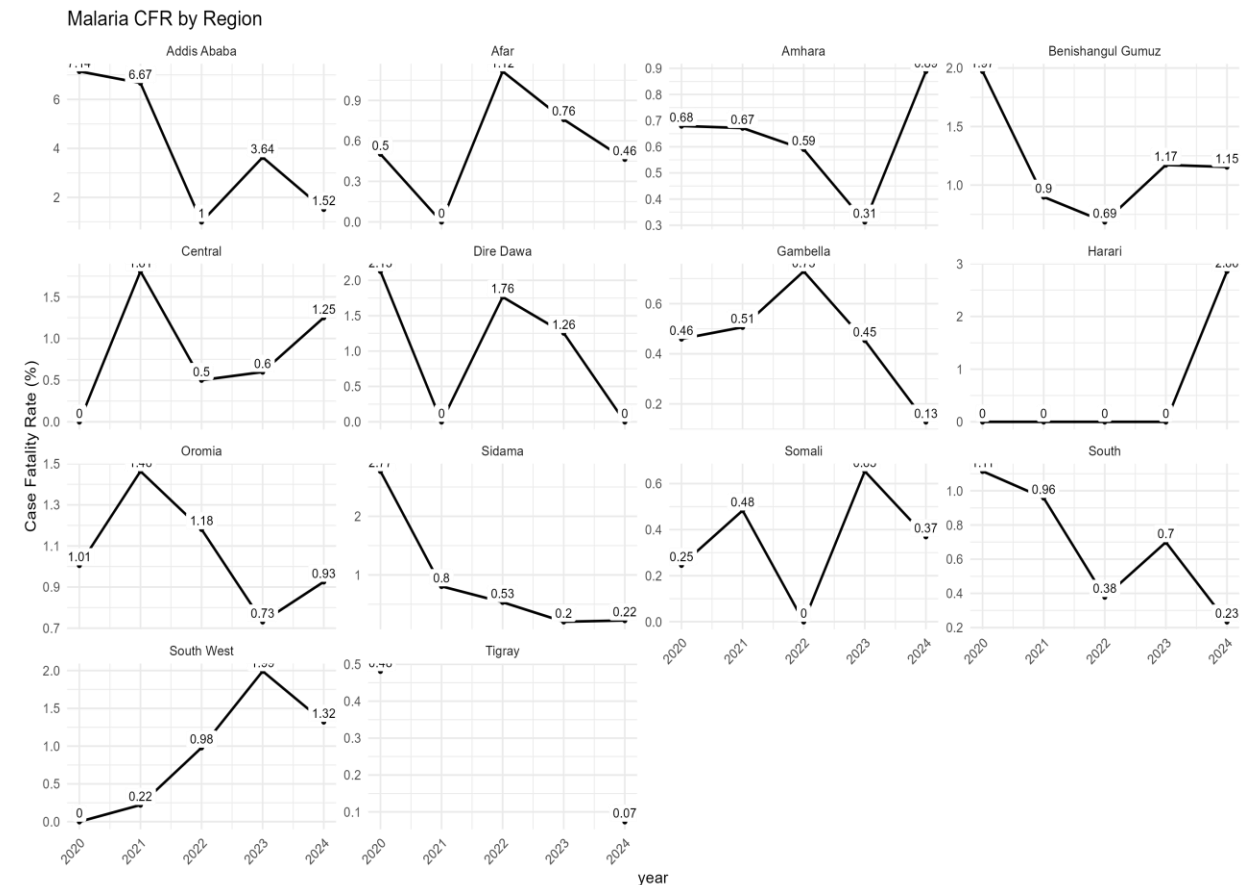
Figure 12: Admission rate among consistently reporting hospitals



4: Malaria Case Fatality Rate

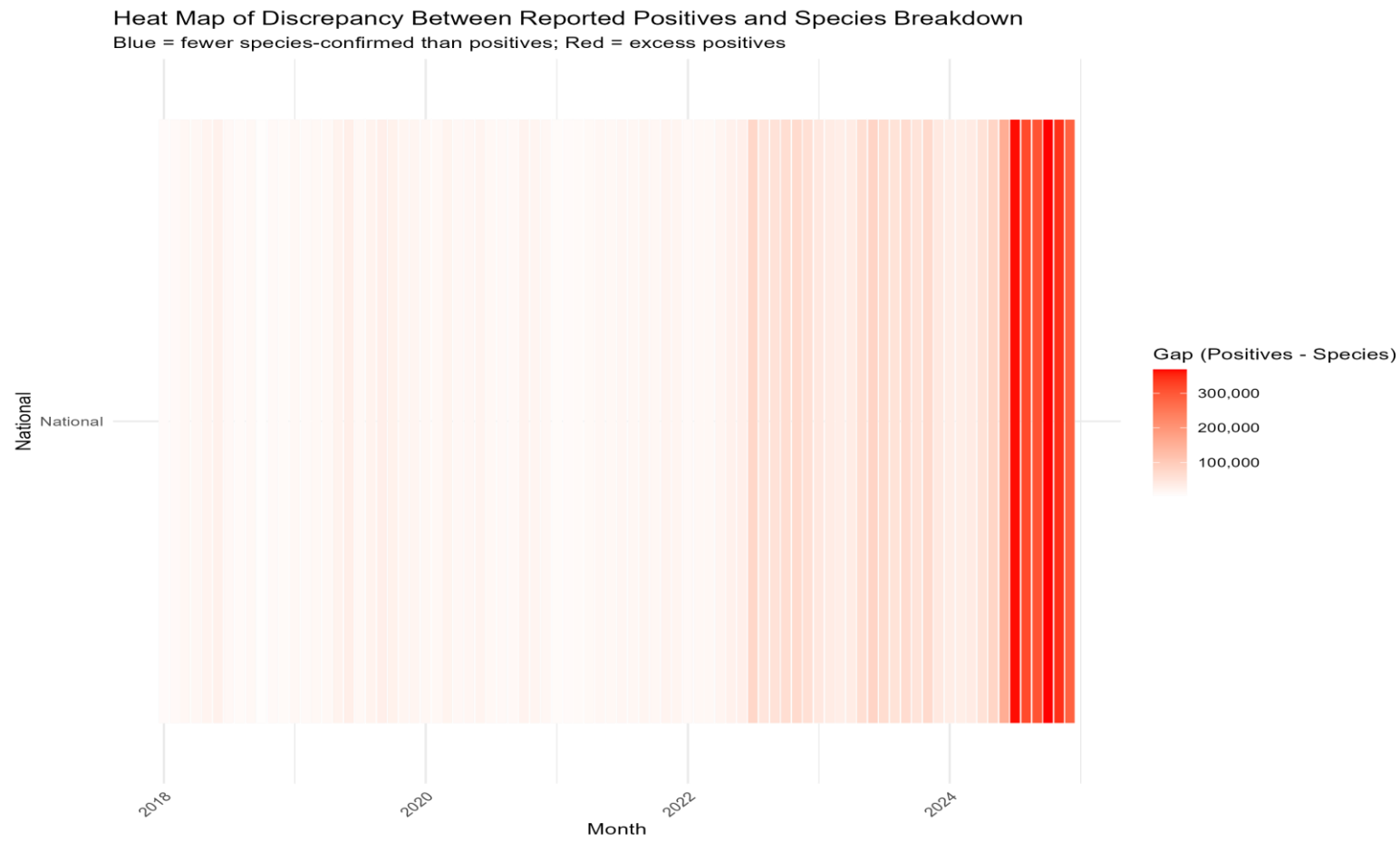
- **Data source: Hospital-level HMIS report**
- Conducted using HMIS data from hospitals that fulfill the consistently reporting facility criteria (9/12)
- Hypothesizing that improved detection of HRP2/3-deleted *Pf* strains at health posts through non-HRP RDTs contributes to fewer severe malaria cases and deaths at the hospital level.
- While 2024 saw stable or lower CFRs in most regions, Amhara, Central, Harari, and Oromia showed a rising trend compared to earlier years.
- Conclusion: While enhanced diagnosis may have supported earlier detection and treatment in some areas, other factors likely influenced the overall CFR trend. Warranted further investigation

Figure 13: CFR by region among consistently reporting hospitals



5. Service vs disease report (National): Internal data quality check

Data source: all HFs HMIS reports



6. Service vs disease report (Regional): Internal data quality check

Data source: all HFs HMIS report

